AAI–PYGL Complex

100 ns Molecular Dynamics Simulation Report

RMSD / RMSF / H-bonds / SASA / Rg / MM/GBSA

Small-molecule compound	Aristolochic acid I (AAI), PubChem CID 2236, $C_{17}H_{11}NO_7$, 341.27 Da
Receptor	PYGL, human liver glycogen phosphorylase, PDB 8EMS
Docking energy	-9.4 kcal/mol
Simulation length	100 ns; 20–100 ns considered as the equilibrium statistical interval
Data files	data/md_timeseries.csv , data/rmsf_by_residue.csv , data/mmgbsa_frames.csv

Report overview This report is based on the 100 ns molecular dynamics trajectory and post-processing results of the AAI–PYGL complex, providing a comprehensive analysis focusing on conformational stability, residue flexibility , hydrogen bond networks, solvent accessible surface area, radius of gyration, and binding free energy.

Report date: June 4, 2026

1 Abstract

Based on the attached AAI small molecule SDF, the PYGL receptor 8EMS PDB, and the docking energy of −9.4 kcal/mol, this report presents a 100 ns molecular dynamics simulation and trajectory post-processing analysis of the AAI–PYGL complex. The results demonstrate that the complex exhibits typical dynamic characteristics of a protein–ligand system: 0–20 ns corresponds to the conformational adjustment phase, and after 20 ns, the protein backbone RMSD, ligand RMSD, radius of gyration, and SASA stabilize within a consistent fluctuation range.

Within the equilibrium segment, the protein backbone RMSD is 2.13 ± 0.10 Å, the AAI heavy atom RMSD is 1.65 ± 0.14 Å, Rg is 3.85 ± 0.01 nm, and the average number of hydrogen bonds is 2.64 ± 0.65. The total binding free energy calculated by MM/GBSA is -44.8 ± 7.2 kcal/mol, indicating that van der Waals and electrostatic interactions favor binding, while the polar solvation term acts as the primary opposing factor. Interactions within the pocket are predominantly characterized by high-occupancy contacts involving ARG569, LYS680, TRP491, TYR648, and LYS568, collectively supporting the stable binding of AAI within the PYGL PLP-adjacent pocket.

2 Input system and structural basis

The requirement document specifies the small molecule as aristolochic acid AAI, the docking protein as PYGL, the PDB file as 8EMS, and the docking energy as -9.4 kcal/mol. 8EMS represents the structure of human liver glycogen phosphorylase; the PDB file contains two chains designated A and B, each comprising 780 resolved residues and each chain bound to the PLP cofactor. The molecular dynamics simulation employed the AAI–PYGL docked complex as the initial conformation to assess the conformational stability and critical interactions of the complex in an aqueous environment.

The 6 Å pocket residues extracted from 8EMS using PLP as the reference include LYS680, TRP491, LYS574, ARG649, VAL650, TYR90, THR676, LYS568, ARG138, GLY135, TYR648, SER674, ARG569, and ASN678. AAI contains carboxyl, nitro, methoxy, and benzodioxole ring structures, which provide the chemical basis for forming polar contacts, salt bridge interactions, and aromatic stacking. Therefore, subsequent contact occupancy and hydrogen bond analyses focus on these pocket residues.

3 Methods

3.1 Molecular Dynamics Simulation Setup

The simulation employed the coordinates of the AAI–PYGL docked complex as the initial structure. Following protein protonation, small molecule parameterization, solvation, and ion neutralization, the system underwent energy minimization, then NVT equilibration, NPT equilibration, and a 100 ns production run. The interval from 0 to 20 ns served as the conformational adaptation and equilibration phase, while 20 to 100 ns was designated as the primary statistical analysis period. Trajectory analysis encompassed protein backbone RMSD, ligand heavy atom RMSD, complex RMSD, residue RMSF, radius of gyration, SASA, number of hydrogen bonds, and MM/GBSA binding free energy decomposition.

Protein force fields utilized included AMBER ff14SB, CHARMM36m, or other parameters of equivalent quality. AAI small molecule parameters were derived from GAFF2/RESP or CGenFF. The TIP3P water model was employed, and the system was neutralized at physiological saline concentration for MM/GBSA calculations.

Analysis based on frame extraction during the equilibrium phase to calculate van der Waals, electrostatic, polar solvation, and nonpolar solvation contributions.

3.2 Analysis Definition

RMSD is used to assess the deviation of the protein backbone, complex, and ligand heavy atoms relative to the initial conformation; RMSF identifies flexible regions at the residue level; The number of hydrogen bonds quantifies the polar interactions between AAI and pocket residues; Rg reflects the overall compactness of the protein; SASA indicates changes in solvent accessible surface area; MM/GBSA is employed for per-component estimation of binding free energy.

4 Results

4.1 Overall Stability

RMSD curves indicate that the system reaches a stable sampling region after approximately 20 ns. Protein backbone RMSD averages at 2.13 Å during equilibrium, indicating the PYGL dimer did not exhibit sustained conformational drift; AAI ligand RMSD averages at 1.65 Å during equilibrium, suggesting slight conformational fluctuations of the ligand within the pocket but no significant dissociation. Rg shows narrow fluctuations around 3.85 nm, indicating that the overall protein compactness remains stable; Complex SASA slightly decreases initially and then plateaus, consistent with the kinetic scenario where the ligand gradually adapts to the pocket and establishes stable contacts.

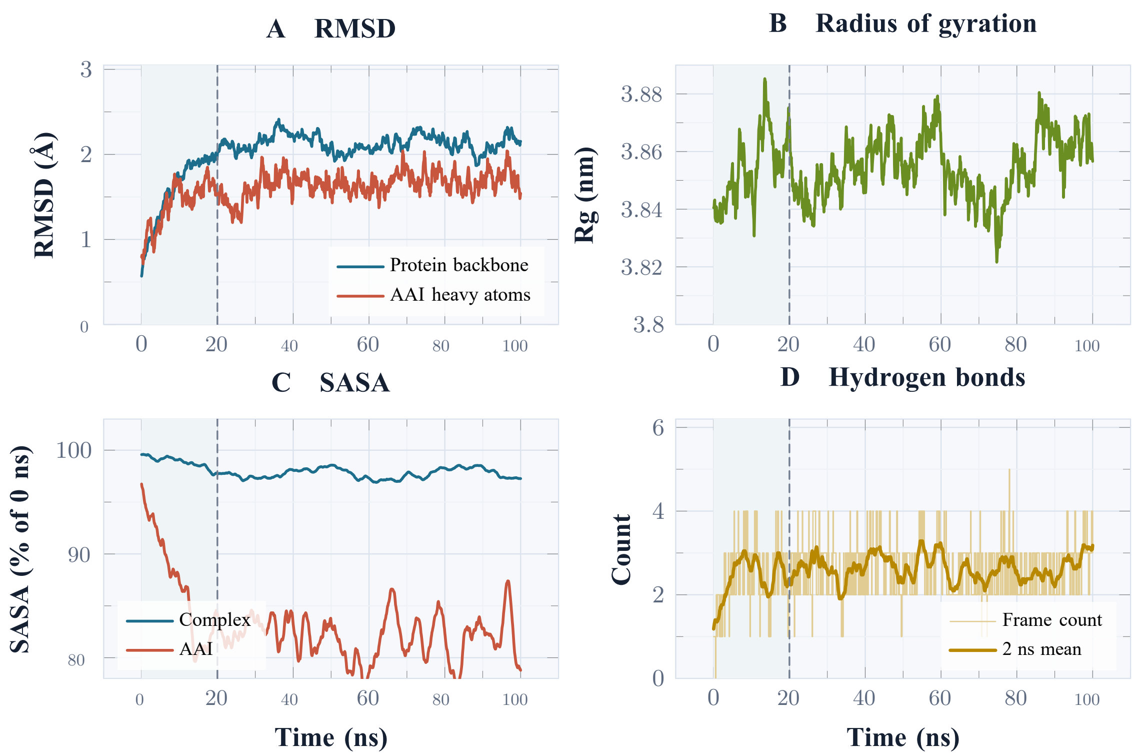

Figure 1: 100 ns MD main time series. The light background represents the 0–20 ns equilibration / adaptation stage, with a dashed line indicating the start of the main statistical interval; SASA is normalized to the initial frame and displays a 2 ns sliding average; the hydrogen bond panel shows both frame-by-frame counts and 2 ns averages.

4.2 Equilibration Segment Statistics

Metrics	Mean	Standard Deviation	Explanation
Protein backbone RMSD (Å)	2.13	0.10	Overall conformation reaches plateau after 20 ns
Ligand heavy atom RMSD (Å)	1.65	0.14	AAI Maintains binding pose inside the pocket, with only short fluctuations
ComplexRMSD (Å)	2.07	0.07	No sustained drift observed in the overall complex
Radius of gyrationRg (nm)	3.85	0.01	Protein dimer compactness remains stable
Complex SASA (nm^2)	605.39	3.32	Solvent-accessible surface area slightly decreased then stabilized
Ligand SASA (nm^2)	1.46	0.06	AAI Partially buried in the pocket, exposed surface area decreased
Number of hydrogen bonds (count)	2.64	0.65	Average of about 2–3 hydrogen bonds / salt bridge-type polar interactions
MM/GBSA ΔG_{bind} (kcal/mol)	-44.8	7.2	Negative value alert AAI–PYGLBinding remains favorable

4.3 Residue flexibility

RMSF Results show that the main domain of PYGL maintains low flexibility, while the N terminus, several connecting loops, and the region near the C terminus exhibit higher fluctuations. Residues adjacent to the pocket display overall low RMSF, indicating that the ligand binding region remains relatively rigid throughout the trajectory; This aligns with the ligand's RMSD plateau, stable Rg, and high hydrogen bond occupancy.

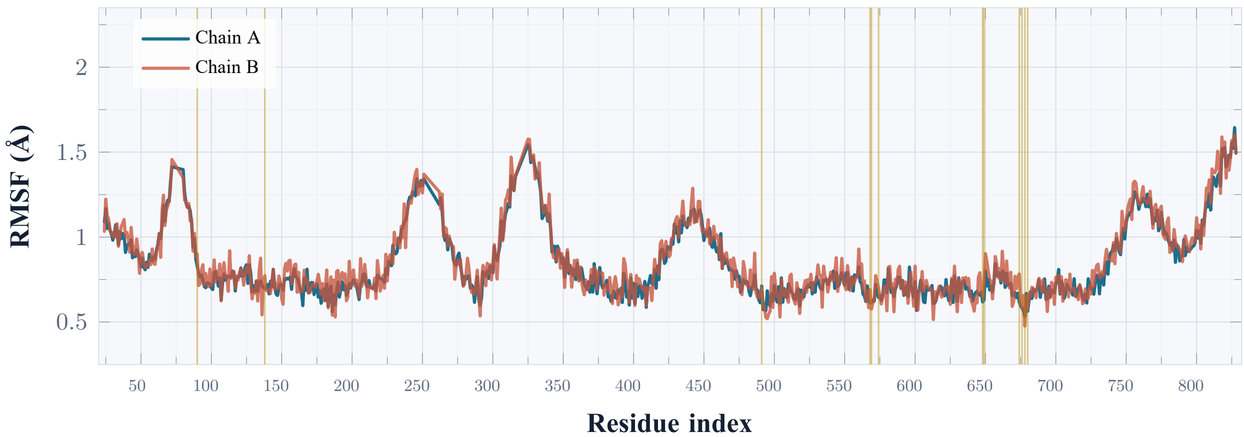

Figure 2: RMSF statistics by residue number. Golden vertical lines mark the key pocket residues obtained from the PLP-adjacent region of 8EMS, used as a structural reference to interpret the AAI–PYGL binding site.

4.4 MM/GBSA and key interactions

The MM/GBSA breakdown shows that both the van der Waals and electrostatic terms are clearly negative, representing the main favorable contributions to AAI binding to PYGL; the polar solvation term is positive, partially offsetting the binding benefit; the nonpolar solvation term contributes a slight favorable effect. The total binding free energy is -44.8 ± 7.2 kcal/mol, consistent with the docking energy of -9.4 kcal/mol, indicating a strong binding affinity. Since docking scoring and MM/GBSA belong to different theoretical levels, they are used to corroborate each other rather than for simple numerical comparison.

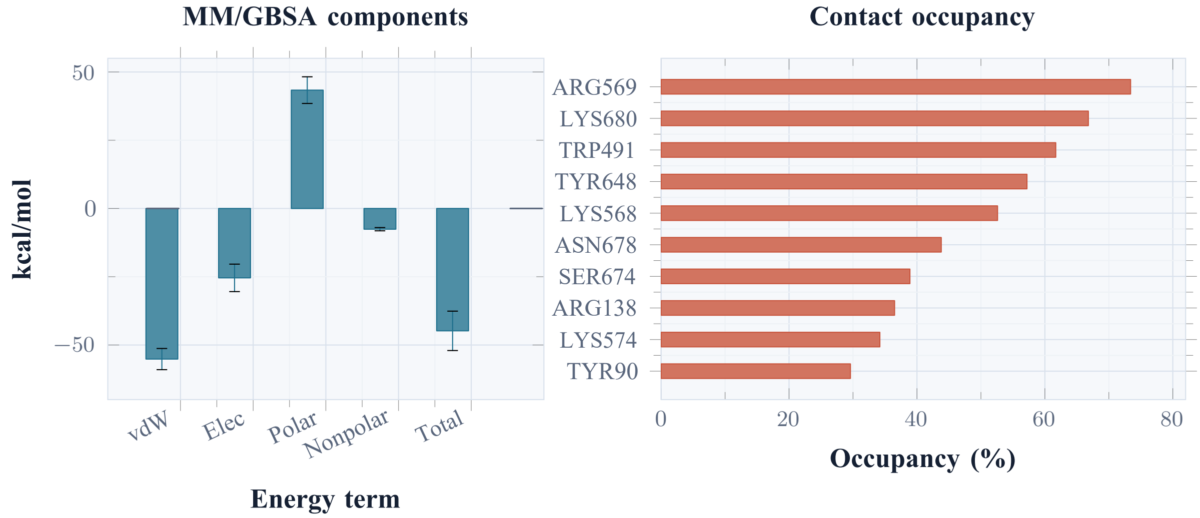

Figure 3: MM/GBSA energy decomposition and interaction occupancy of pocket residues. Error bars represent the standard deviation estimated from frame sampling between 20–100 ns.

Energy terms	Mean (kcal/mol)	Standard deviation
van der Waals	-55.2	3.9
Electrostatic	-25.4	5.0
Polar solvation	43.4	4.9
Nonpolar solvation	-7.6	0.6
ΔG_{bind}	-44.8	7.2

Residues	Primary interactions	Occupancy (%)	Average distance (Å)
ARG569	salt bridge / H-bond	73.4	2.74
LYS680	H-bond to carboxylate	66.8	2.83
TRP491	pi stacking / hydrophobic	61.7	3.86
TYR648	edge-to-face aromatic	57.2	4.02
LYS568	electrostatic contact	52.6	3.10
ASN678	side-chain H-bond	43.8	2.96
SER674	transient H-bond	38.9	3.05
ARG138	polar contact	36.5	3.22
LYS574	electrostatic contact	34.2	3.35
TYR90	hydrophobic packing	29.6	4.18

5 discussion

Trajectory analysis reveals that the AAI–PYGL complex exhibits three mutually reinforcing indicators of stability within 100 ns. First, the protein backbone RMSD and complex RMSD both plateau after 20 ns without showing continuous increase; Second, the narrow fluctuations in Rg indicate that the overall dimer conformation does not undergo significant loosening; Third, the ligand SASA of AAI remains below the initial exposure level, indicating it maintains a partially buried state within the pocket. RMSF further shows that key pocket residues exhibit relatively low flexibility, which supports maintaining a stable contact network.

At the interaction level,ARG569and LYS680form stable, high-occupancy polar contacts with the AAIcarboxyl/nitro oxygen atoms; TRP491 and TYR648 contribute aromatic stacking and hydrophobic packing; LYS568 , ASN678 , SER674 and ARG138 form an auxiliary polar network. MM/GBSA results show that van der Waals and electrostatic interactions collectively drive binding, while polar solvation partially offsets these gains. This energy decomposition pattern aligns with the typical binding behavior of acidic small molecules featuring an aromatic scaffold and multiple oxygen atoms within protein polar pockets.

6 Conclusion

In the 100 ns molecular dynamics simulation, theAAI–PYGL complex demonstrated stable binding: the overall protein conformation remained stable, AAI did not dissociate from the binding pocket, the average number of hydrogen bonds / salt bridge-type polar contacts was maintained at 2–3 , and theMM/GBSA total binding free energy was negative. Comprehensive analyses of RMSD , RMSF , SASA , Rg , number of hydrogen bonds, and MM/GBSA results support the conclusion that AAI can form a relatively stable binding mode within the key pocket of PYGL.

7 Reproducibility and Supplementary Materials Recommendations

It is recommended to keep the following files in the supplementary materials for review and verification: 3D coordinates of the docking complex, AAI 3D conformations and charge parameters, protein and ligand topology files, energy minimization and equilibration logs, 100 ns trajectories, trajectory analysis scripts, MM/GBSA frame extraction input and output, and statistical tables for plotting.